



MEMORANDUM

To: Professors Ageenko and Schuckers

From: Ayia Boutanassa, Brady Dunn, Bryce Jackson, Eric Testen, Garrison High

Date: April 27, 2026

RE: Project Proposal: Housing Unaffordability and Displacement Risk in Charlotte, NC

Overview

This project investigates the primary drivers of housing unaffordability in Charlotte and identifies neighborhoods at high risk of displacement. A secondary analysis will examine whether these high-risk neighborhoods were historically redlined by the Homeowners' Loan Corporation (HOLC), providing evidence of long-term impacts.

Data Sources

Our datasets include: (1) the Charlotte Area Quality of Life (QoL) Explorer, accessed by the City of Charlotte's ArcGIS REST API. This provides socioeconomic, demographic, housing, and infrastructure variables by Neighborhood Profile Areas (NPA); (2) the 'Vulnerability to Displacement by NPA' dataset from the Charlotte Open Data Portal which contains the classification target; and (3) the 'Mapping Inequality' dataset from the University of Richmond, containing historical HOLC redlining grades (A-D) for Charlotte neighborhoods.

Modelling

Our Multiple Linear Regression model will predict a composite housing cost burden index. This target variable will be derived from rental cost and home sales price relative to house income, weighted by homeownership rate. Our classification model will predict binary displacement risk. We will experiment with using Logistic Regression, kNN, and Random Forest. Features considered will involve socioeconomic status (income, FNS, health insurance, employment), neighborhood characteristics (housing age, code violations, density, foreclosures), demographics, safety, and infrastructure accessibility.

Expected Outcomes

Model outputs will identify neighborhood characteristics strongly associated with unaffordability and displacement risk. Flagged neighborhoods will be overlaid onto historical HOLC redlining boundaries to assess whether discriminatory housing practices from 1935 to 1968 continue to shape housing outcomes in Charlotte today.